

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	26 June 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements – Metro Ticket Generating System

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Role Management	Authenticate users using the existing ServiceNow Users (sys _ user) table. Assign Metro Passenger role to users. Assign Metro Administrator role to administrators. Provide role-based access to application modules.
FR-2	Metro Ticket Booking	Passengers can select Source Station and Destination Station . Select journey date and ticket type (Single/Return). Create a metro ticket booking request. Validate station selection before booking.
FR-3	Fare Calculation & Payment	Automatically calculate fare based on source and destination stations. Display fare before payment. Allow online payment through supported payment methods. Update booking status after successful payment.
FR-4	QR Ticket Generation & Validation	Generate a unique QR-code metro ticket after successful payment. Store ticket details in the Metro Ticket table. Allow passengers to view and download their tickets. Validate QR tickets at metro entry and exit gates.
FR-5	Metro Administration	Administrators can add, edit, and delete metro stations. Manage metro routes and station connections. Configure fare rules between stations. View all passenger bookings and ticket history.
FR-6	Workflow Automation & Notifications	Trigger a Flow Designer workflow after ticket booking. Send booking confirmation email. Send payment confirmation notification. Send ticket cancellation notification (if applicable). Generate audit logs for every ticket transaction.
FR-7	Reports & Dashboard	Passengers can view booking history and active tickets. Administrators can view dashboards showing ticket bookings, revenue, passenger statistics, and payment reports. Generate downloadable reports for analysis.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

Non-Functional Requirements – Metro Ticket Generating System

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide an intuitive and user-friendly interface for passengers and administrators, enabling easy ticket booking, fare viewing, and ticket management.
NFR-2	Security	Access shall be controlled using ServiceNow roles. Passengers may access only their own ticket bookings, while administrators may manage stations, routes, fares, and bookings based on their assigned roles.
NFR-3	Reliability	The application shall accurately store ticket bookings, payment details, and QR ticket information while ensuring workflow execution without data loss.
NFR-4	Performance	Ticket booking, fare calculation, QR ticket generation, payment processing, and email notifications shall be completed with minimal delay under normal system usage.
NFR-5	Availability	The Metro Ticket Generating System shall be available whenever the ServiceNow instance is operational, allowing passengers to book tickets at any time.
NFR-6	Scalability	The application shall support increasing numbers of passengers, metro stations, routes, ticket bookings, and concurrent users without requiring significant architectural changes.
NFR-7	Maintainability	The application shall be modular and easy to maintain, allowing administrators to update stations, routes, fare rules, and workflows with minimal effort.
NFR-8	Compatibility	The application shall function correctly on desktop browsers, tablets, and mobile devices supported by the ServiceNow platform.
NFR-9	Data Integrity	The system shall ensure accurate fare calculations, prevent duplicate ticket generation, and maintain consistency of booking and payment records.
NFR-10	Audit & Logging	The system shall maintain logs of ticket bookings, payments, cancellations, and administrative actions to support monitoring and troubleshooting.